2 Limits

2.1 **The Limit of a Function**

2.1.1 **Two-sided limits.**

The notion of a limit permeates much of our work in calculus. We begin with the intuitive introduction to the limits. We use limits to describe the way a function varies. Some functions vary continuously; small changes in x produce only small changes in f (x) .

Definition 17 Let f be a function defined on an open interval containing a, with the possible exception of a itself. Then the limit of f (x) as x approaches a is the number L, written

$$\lim_{x \to a} f(x) = L$$

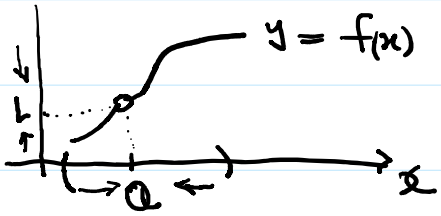
if f (x) can be made abitrarly close to L (as close as we like) by taking x to be sufficiently close to a (on either side of a) but not equal to a.

An alternative notation for

use

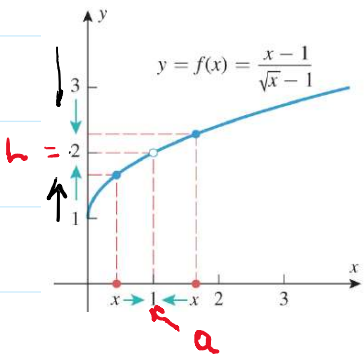
$$\lim_{x \to a} f(x) = L$$

is $f(x) \to L$ as $x \to a$.



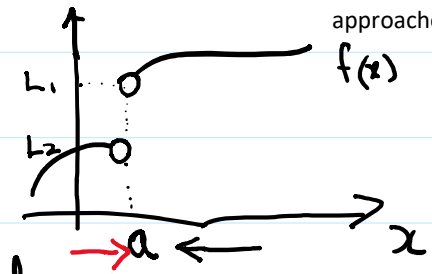
Example : Use the graph to evaluate $\lim_{x \to 1} f(x)$ if

$$f(x) = \frac{x-1}{\sqrt{x}-1}$$



$$\lim_{x \to 1} \frac{x-1}{\sqrt{x}-1} = 2$$

We note that, in finding the limit, we are only interested in what happens to f (x), when x approaches 1, not what happens when x = 1.



$\lim_{x \to a} f(x)$ does not exist

right hand limit

$$\lim_{x \to a^+} f(x) = L_1$$

left hand limit

$$\lim_{x \to a^-} f(x) = L_2$$

Limits and Continuity...

Each of the two limits above is refered to as **a one sided limit** whereas $\lim_{x\to a} f(x)$ is a **two sided limit.**

In general, there is no guarantee that a function f will have a two-sided limit at a given point a; that is, the values of $f(x)$ may not get closer and closer to any single real number L as $x \to a$. In this case we say that

$$\lim_{x\to a^+} f(x) = L_1 \neq L_2 = \lim_{x\to a^-} f(x)$$

$$\lim_{x\to a} f(x) \quad \textbf{does not exist}$$

Similarly, the values of $f(x)$ may not get closer and closer to a single real number L as $x \to a^+$ or as $x \to a^-$. In these cases we say that

$$\lim_{x\to a^+} f(x) \quad \textbf{does not exist}$$

or that

$$\lim_{x\to a^-} f(x) \quad \textbf{does not exist}$$

Limits and Continuity...

Relationship between one-sided and two-sided limits

The **two-sided limit** of a function $f(x)$ exists at a if and only if both of the **one-sided limits** exist at a and have the same value; that is,

$$\lim_{x\to a} f(x) = L \quad \text{if and only if} \quad \lim_{x\to a^+} f(x) = L = \lim_{x\to a^-} f(x)$$

Thus the (two sided) limit exists if and only if the one-sided limits exist and are equal.

Example

Find $\lim_{x\to 0} \frac{|x|}{x}$.

Recall : $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

$- \quad + \quad 0$

$$\lim_{x\to 0^+} \frac{x}{x} = \lim_{x\to 0^+} 1 = 1$$

$$\lim_{x\to 0^-} \frac{-x}{x} = \lim_{x\to 0^-} -1 = -1$$

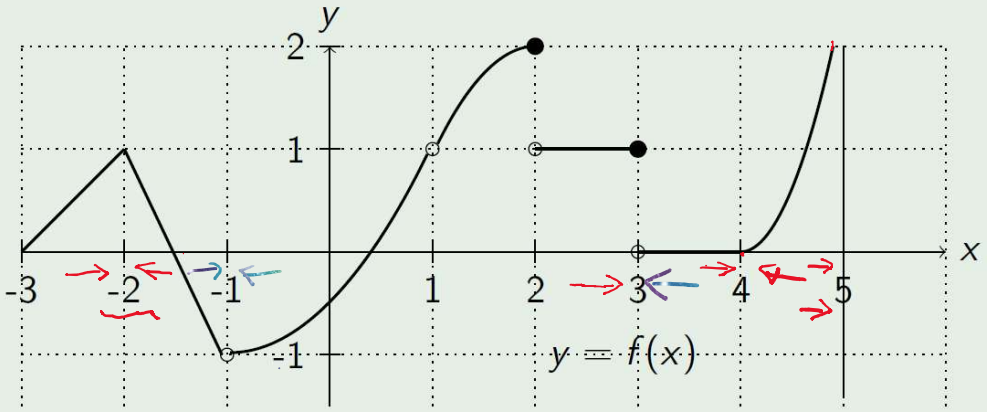
$\lim_{x\to 0} \frac{|x|}{x}$ does not exist

$\because \lim_{x\to 0^+} \frac{|x|}{x} \neq \lim_{x\to 0^-} \frac{|x|}{x}$

Limits and Continuity...

Example

Use the figure below to find the following limits:



(a) $\lim_{x\to -2} f(x)$ $=1$

(b) $\lim_{x\to -1} f(x)$ $=-1$

(c) $\lim_{x\to 1} f(x)$ $=1$

(d) $\lim_{x\to 2^-} f(x)$ $=2$

(e) $\lim_{x\to 2^+} f(x)$ $=1$

(f) $\lim_{x\to 2} f(x)$ does not exist

(g) $\lim_{x\to 3^-} f(x)$ $=1$

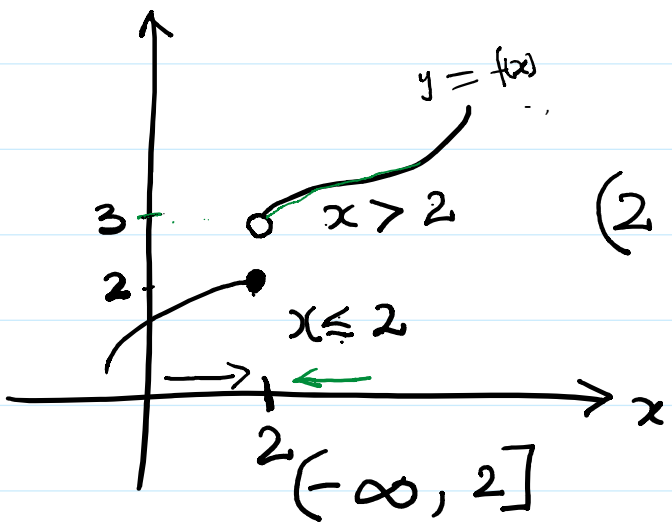
(h) $\lim_{x\to 3^+} f(x)$ $=0$

(i) $\lim_{x\to 4^-} f(x)$ $=0$

(j) $\lim_{x\to 4^+} f(x)$ $=0$

(k) $\lim_{x\to 4} f(x)$ $=0$

(l) $\lim_{x\to 5^-} f(x)$ $=-\infty$



$$\lim_{x\to 2^-} f(x) = 2$$

$$\lim_{x\to 2^+} f(x) = 3$$

$$\lim_{x\to 2^-} f(x) \neq \lim_{x\to 2^+} f(x)$$

$\therefore \lim f(x)$

1 2 $(-\infty, 2]$

$x \to 2^-$ | $x \to 2^+$

$\therefore \lim_{x \to 2} f(x)$ DNE